

Review Radar

Streaming review intelligence: which products are declining, and what are people complaining about?

Amazon Reviews 2023 · All_Beauty · 701,528 reviews

Philip Bergman · BIU 8688697201, Big Data and AI · final project (solo)

The question, and who asks it

Which products experienced a sustained decline in customer ratings, and which complaint themes increased during that decline?

Asked by a category manager. Fixed before the analysis was run.

701,528

reviews

112,590

catalogue products

2000 → 2023

2000-11-01 to 2023-09-09

The answer, in one line

The completed result identifies sustained rating declines; aspect-level explanation is preliminary.

139 of 502

eligible products raised an alert during 2020–2023, under a rule frozen beforehand

0.80 / month

false alerts on placebo data, against a budget of one

18.6%

of declines its own size are detected, so this is a floor, not a census

Hero: B00RPJZMUM — `hard_to_use` and `not_as_described` give way to `does_not_work` (“motor stopped working”).

An illustration over 18 baseline and 3 recent labelled mentions, not an estimate. The decline claim stands; the theme claim is illustrative.

The data, and which V's it exercises

VOLUME

311 MB → 96.5 MB

bronze Parquet, 3.2× smaller · 0.54 GB raw JSONL ·
693,547 ES docs · 345,418 vectors

VELOCITY

232k–484k rec/s

producer throughput, a range · the stream replay is paced to
a frozen 3,000 rec/s, measured 2,999

VARIETY

free text + free-form JSON

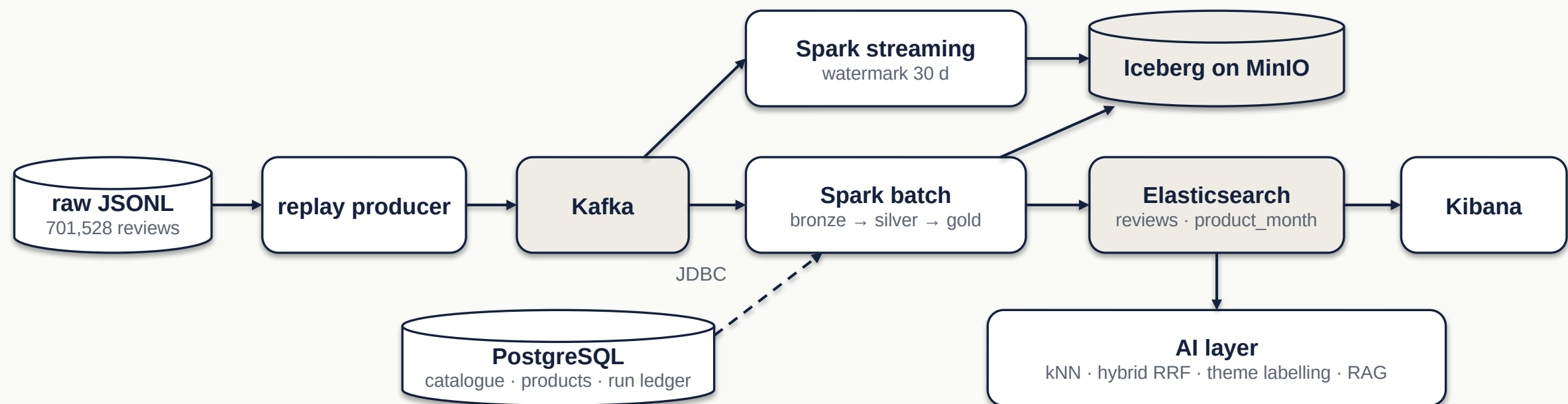
a details dict whose keys collide case-insensitively · price as
null, “9.99”, “\$9.99” or a range · Postgres catalogue joined
over JDBC

VERACITY

6,139 collision groups

over 13,415 rows: 6,138 byte-exact, 1 conflicting on
helpful_vote, 0 disagreeing on rating → 7,276 rows removed

Architecture and data flow



EXACTLY-ONCE, AS A FIGURE

120,000 records expected	89,964 rows before the SIGKILL	120,000 rows after the restart	0 duplicate (partition, offset) pairs
------------------------------------	--	--	---

The AI capability, and why this one

“BM25 does not consider the semantic meaning of the query terms or the documents.”

the course, deck 5

345,418

MiniLM-L6-v2 vectors over the ≥ 20 -word reviews, indexed beside BM25 in Elasticsearch

0.96

ANN recall@10 against exact cosine on the frozen query set

H-E1 not held

kNN alone does not beat BM25 — client-side RRF hybrid is where the gain is

Hybrid: two rankings fused client-side by reciprocal rank, $\Sigma 1/(60 + \text{rank})$.

Themes: a frozen ten-theme complaint taxonomy, labelled per review by a local qwen3:8b.

RAG: answers with claim-level citations over the same retriever; 30 frozen questions.

Live demo

Ten moves, one kernel · 4:40

01 The stack is real

02 Paced replay + streaming projection

03 Silver's reconciliation identity

04 Every run this data went through

05 Decline candidates (Kibana)

06 Hybrid decomposition

07 Complaint themes for the top candidate

08 The evaluation table

09 RAG answers, claim by claim

10 The stream's own gate

How we avoided fooling ourselves

1

Temporal holdout, protocol freeze

The decline rule was developed on pre-2020 data and sealed in one commit before a single 2020–2023 point was evaluated.

2

Reproducibility is separate from outcome

12 gates PASS on re-derivation; two re-derive silver and gold in pandas with no shared code (13 and 11 checks, 0 mismatched fields).

3

Bars set before the measurement, never moved

When the labeller dropped from a hosted model to a local 8B model, 0.70 stayed 0.70. The audit set was opened once.

4

Every number joins back to a run

A run_id in every Iceberg snapshot, ES doc and eval artefact; the lineage gate walks 101 links.

Results, zoomed out

139 of 502

eligible products alerted in the 2020–2023 holdout (27.7%)
— 148 alerts, 148 episodes

425 episodes

over the whole spine — closed by recovery 207, gap 165,
end of data 53

0.186 vs 0.80

detection power at the 0.3★ step [0.168, 0.205] against the
pre-registered bar — a published FAIL

0.4583 vs 0.70

theme macro-F1 [0.3647, 0.5223] — a published FAIL, but
above the star-only floor 0.2968; MLlib baseline 0.3899

An alert is unlikely to be noise. The 139 are a floor, not a census — and an alert is not causal proof.

Trade-offs, and what we declined

THREE TRADE-OFFS

Iceberg is not a course technology. One mention in ~860 slides; conceded up front, earned on mechanism.

A local qwen3:8b instead of a hosted model. The 0.70 bar deliberately unmoved, so the constraint shows as a reported miss.

Public but sensitive review text. Only title and text cross the wire to the model, never user_ids.

FOUR DECLINES, SAID OUT LOUD

Kafka Connect Elasticsearch sink — dropped for time. The schedule, not the merits.

Single-node HDFS — declined on resources, and MinIO is not equivalent. No block replication, no atomic rename; hence Postgres holds the Iceberg catalogue.

Sqoop and Pig — on the course's own slide. “Apache Sqoop moved into the Attic in June 2021”; Spark SQL does Pig's transform.

Oozie — an opinion, and I own it as one. Spark and the run ledger fill the role, but Oozie is not retired.